

**We acknowledge the use of OpenAI's ChatGPT as a supportive tool in drafting and refining portions of the text.*

LAND 430 / LAND 509 Urban Environmental Analytics Syllabus 2026/27

W1

University of British Columbia

1 COURSE INFORMATION

Course Title	Course Code Number	Credit Value
Urban Environmental Analytics	LAND 430 / LAND 509	3

PREREQUISITES

An introductory course in environmental data analysis or statistics is essential to understanding the course content. Additional for LAND 430: Third-year standing or permission of instructor.

COURSE DESCRIPTION

Environmental data are increasingly used to guide sustainability decisions, ecosystem monitoring, and climate adaptation strategies. This course introduces students to applied environmental analytics using statistical modelling, visualization methods, geospatial analysis, and environmental monitoring datasets.

2 CONTACT

Course Instructor	Contact Details	Office Location	Office Hours
Dr. Elena Brooks	Discussion board preferred Email: elena.brooks@ubc.ca Response time within 48 hours	FSC 2331	Mon 11-12 Wed 14-15

3 COURSE STRUCTURE

LECTURES

Tuesday, Thursday 12:30 to 13:50 Location: FSC 1220

LABS SESSIONS AND ASSIGNMENTS

Lab assignments apply analytical concepts using environmental datasets and R.

4 LEARNING OUTCOMES

1. Apply statistical and environmental modelling techniques.
2. Visualize and interpret environmental monitoring data.
3. Evaluate analytical outputs and model assumptions.

5 LEARNING ACTIVITIES

Take notes during lectures.
Participate in labs and discussion boards.
Complete analytical exercises and readings.

6 ASSESSMENTS OF LEARNING

Assignments 20%
Participation 10%
Midterm 30%
Final Exam 40%

7 LEARNING MATERIALS

REQUIRED READING MATERIALS

Brooks, E. & Ramirez, J. 2022. *Environmental Analytics and Urban Systems*. West Coast Academic Press.

Foster, P. 2021. *Applied Data Visualization with R*. Northern Research Publishing.

8 SCHEDULE OF TOPICS

Week	Week day	Dates	Computer Labs/Assignments/Exams
1	T Th	Sept 8 - lecture Sept 10 - lecture	Assignment 1: Intro to R and environmental datasets
2	T Th	Sept 15 - lecture Sept 17 - lecture	Assignment 2: Exploratory data analysis
3	T Th	Sept 22 - lecture Sept 24 - lecture	Assignment 2 (cont'd): Data visualization in R
4	T Th	Sept 29 - lecture Oct 1 - lecture	Assignment 3: Regression modelling
5	T Th	Oct 6 - lecture Oct 8 - lecture	Assignment 4: Model diagnostics
6	T Th	Oct 13 - lecture Oct 15 - lecture	Review session – Location: ANGU 347
7	T Th	Oct 20 - lecture Oct 22 - lecture	MIDTERM EXAM – Location: DMP 110
8	T Th	Oct 27 - lecture Oct 29 - lecture	Assignment 6: Environmental prediction
9	T Th	Nov 3 - lecture Nov 5 - lecture	Assignment 7: Climate datasets
10	T Th	Nov 10 - no class Nov 12 - no class	Assignment 7 (cont'd)
11	T Th	Nov 17 - lecture Nov 19 - lecture	Assignment 8: Scenario modelling
12	T Th	Nov 24 - lecture Nov 26 - lecture	Assignment 9: Urban ecosystem indicators
13	T Th	Dec 1 – guest lecture Dec 3 – final lecture	Review session – Location: ANGU 347
		December 8 to 20: Final Exam Period	Final Exam: Date and location to be announced

9 UNIVERSITY POLICIES

Students are expected to uphold academic integrity standards and communicate respectfully during all course activities.

10 OTHER COURSE POLICIES

All course materials are the intellectual property of the course instructor or licensed for educational use.

11 ACKNOWLEDGEMENT

UBC's Point Grey Campus is located on the traditional, ancestral, and unceded territory of the xwm̓θkw̓y̓m̓ (Musqueam) people.